

Faculty of Computing

Year 2 Semester 1 (2026)

IT2011 - Artificial Intelligence and Machine Learning

Progress Review I - Data Preprocessing and EDA

You are required to attend the Progress Review I Viva, which focuses on your Data Cleaning, Preprocessing, and Exploratory Data Analysis (EDA), applied to your assigned dataset.

Progress Review – Viva Details

- Mode: In-class (during your regular lab session)
- Duration: Approximately 20 minutes per group
- Format: Viva-style Q&A + brief walkthrough of your notebook/script

Individual Requirement (20 marks per student):

- Each group member must handle and present at least one preprocessing technique, such as:
 - Handling missing data
 - Encoding categorical variables
 - Outlier removal
 - Normalization / scaling
 - Feature engineering (feature selection and dimension reduction), etc.
- Each member must:
 - Clearly explain the selected technique
 - Justify why it was needed for the assigned dataset
 - Show the implementation (code/output)
 - Present at least one EDA visualization (e.g., histogram, boxplot, correlation matrix) and interpret it

Students who do not present or fail to show individual contribution will receive zero for the individual component.

Group Requirement (5 marks shared):

- Submit a combined preprocessing pipeline that reflects proper integration of each member's work
- Demonstrate logical flow and collaboration
- Clearly commented and organized code

Please ensure each team member is prepared with their section of the code and visual output. This is a critical milestone in your assignment and contributes directly to your final evaluation.

Evaluation Details

- Format: Viva
- Duration: 15 minutes per group
- Mode: In-class during Lab Sessions

Repository Layout for Group Deliverable folder (Upload a Zip file to Courseweb)

Group_ID/

- README.md — Overview of the project, assigned dataset details, group member roles, how to run code
- data/raw/ — Assigned dataset(s) as provided
- data/external/ — Any external reference datasets (if used)
- notebooks/IT_Number_Preprocessing_technique.ipynb — one per member (Member 1–6), named with IT Number & technique
- group_pipeline.ipynb — Integrated pipeline (combined work)
- results/eda_visualizations/ — Plots & charts (PNG/JPEG)
- results/logs/ — Any logs from execution (optional)
- results/outputs/ — Final processed dataset / features